

C Assignment

1. Find sum of two integers

```
#include <stdio.h>
int main() {
    int a,b;
    printf("Enter the first number : ");
    scanf("%d", &a);
    printf("Enter the second number : ");
    scanf("%d", &b);
    printf("Sum = %d \n", a + b);
    return 0;
}
```

```
Enter the first number : 10
Enter the second number : 5
Sum = 15
```

2. Find out the area and perimeter of a rectangle

```
#include<stdio.h>
#include<math.h>
int main() {
    int a,b,c;
    printf("Enter the first side : ");
    scanf("%d", &a);
    printf("Enter the second side : ");
    scanf("%d", &b);

    printf("Perimeter = %d \n", 2*(a+b));
    printf("Area = %d\n", a*b);
}
```

```
Enter the first side : 3
Enter the second side : 4
Perimeter = 14
Area = 12
```

3. Input three decimal numbers and find their sum and average.

```
#include<stdio.h>
int main() {
    double a,b,c;
```

```

printf("Enter the first number : ");
scanf("%lf", &a);
printf("Enter the second number : ");
scanf("%lf", &b);
printf("Enter the third number : ");
scanf("%lf", &c);

printf("Sum = %f \n" , a + b + c);
printf("Average = %f \n" , (a + b + c)/3);
return 0;
}

```

```

Enter the first number : 2.57
Enter the second number : 3.54
Enter the third number : 5.25
Sum = 11.360000
Average = 3.786667

```

4. Input two numbers and swap them.

```

#include <stdio.h>
int main() {
    int a ,b;
    printf("Enter the first number : ");
    scanf("%d", &a);
    printf("Enter the second number : ");
    scanf("%d", &b);

    printf(" a = %d \n", a);
    printf(" b = %d \n", b);

    a = a + b;
    b = a - b;
    a = a - b;

    printf("\n Swapping... \n \n");

    printf(" a = %d \n", a);
    printf(" b = %d \n", b);
    return 0;
}

```

```

Enter the first number : 10
Enter the second number : 15
a = 10
b = 15

```

Swapping...

```
a = 15  
b = 10
```

5. Convert temperature in Celsius to temperature in Fahrenheit

```
#include <stdio.h>  
int main() {  
    double c;  
    printf("Enter temperature in Celcius :");  
    scanf("%lf", &c);  
  
    double f = ((c/5)*9) + 32;  
    printf("Temperature in Ferenheit = %.3f", f);  
    return 0;  
}
```

```
Enter temperature in Celcius :75  
Temperature in Ferenheit = 167.000
```

6. Implement simple arithmetic calculator using user defined functions for each operation (addition, subtraction, multiplication, division, modulus, exponent).

```
#include<stdio.h>  
int main() {  
    int n, a, b;  
    printf("Operation -> \n ");  
    printf("1 for +, \n");  
    printf("2 for -, \n");  
    printf("3 for *, \n");  
    printf("4 for //, \n");  
    printf("5 for %, \n");  
    printf("6 for **, \n");  
    printf("0 to exit. \n");  
    while (1 == 1) {  
        printf("Enter the first number :");  
        scanf("%d", &a);  
        printf("Enter the second number :");  
        scanf("%d", &b);  
        printf("Enter the operation :");  
        scanf("%d", &n);  
        if (n == 1) {  
            printf("%d \n", a + b);  
        } else if (n == 2) {  
            printf("%d \n", a - b);  
        } else if (n == 3) {  
            printf("%d \n", a * b);  
        } else if (n == 4) {  
            printf("%d \n", a / b);  
        } else if (n == 5) {  
            printf("%d \n", a % b);  
        } else if (n == 6) {  
            printf("%d \n", a ** b);  
        }  
    }
```

```

        printf("%d \n", a * b);
    } else if (n == 4) {
        if ( b != 0) {
            printf("%f \n", a / (b * 1.0));
        } else {
            printf("Zero division !! \n");
        }
    } else if (n == 5) {
        if ( b != 0) {
            printf("%f \n", a % b);
        } else {
            printf("Zero division !! \n");
        }
    } else if (n == 6) {
        printf("%lf \n", pow(a,b));
    } else if (n == 0) {
        break;
    } else {
        printf("Please enter a valid operation. \n");
    }
}
}

```

Operation ->

```

1 for +,
2 for -,
3 for *,
4 for //,
5 for %,
6 for **,
0 to exit.
Enter the first number :45
Enter the second number :8
Enter the operation :4
5.625000
Enter the first number :0
Enter the second number :5
Enter the operation :0

```

7. Check an integer whether it is odd or even

```

#include<stdio.h>
int main() {
    int a;
    printf("Enter the number: ");
    scanf("%d", &a);
    if (a%2 == 0) {
        printf("%d is Even", a);
    }
}

```

```

    } else if (a%2 == 1) {
        printf("%d is Odd", a);
    }
    return 0;
}

```

Enter the number: 65
65 is Odd

8. Find greatest of 3 integers.

```

#include <stdio.h>
int main() {
    int a,b,c;
    printf("Enter the first number : ");
    scanf("%d", &a);
    printf("Enter the second number : ");
    scanf("%d", &b);
    printf("Enter the third number : ");
    scanf("%d", &c);

    int m;
    m = (a > b) ? ((a > c) ? a : c) : ((b > c) ? b : c);
    printf("Greatest -> %d \n", m);
    return 0;
}

```

Enter the first number : 3
Enter the second number : 7
Enter the third number : 9
Greatest -> 9

9. Find 2nd greatest of 3 integers

```

#include <stdio.h>
int main() {
    int a,b,c;
    printf("Enter the first number : ");
    scanf("%d", &a);
    printf("Enter the second number : ");
    scanf("%d", &b);
    printf("Enter the third number : ");
    scanf("%d", &c);

    int m;
    m = (a > b) ? ((a > c) ? ((b > c) ? b : c) : a) : ((b > c) ? ((a > c)

```

```

? a : c) : b);
    printf("2nd GREATEST -> %d \n", m);
    return 0;
}

```

```

Enter the first number : 6
Enter the second number : 8
Enter the third number : 9
2nd GREATEST -> 8

```

10. Input a number and find its absolute value.

```

#include<stdio.h>
int main() {
    int a;
    printf("ENTER a number :");
    scanf("%d", &a);
    printf("|%d| = %d", a, ((a >= 0)? a : -a));
    return 0;
}

```

```

ENTER a number :-100
|-100| = 100

```

12. Solve a given quadratic equation. (Without imaginary roots).

```

#include <stdio.h>
int main() {
    double a,b,c;
    printf("Enter the first coeff. : ");
    scanf("%lf", &a);
    printf("Enter the second coeff. : ");
    scanf("%lf", &b);
    printf("Enter the third coeff. : ");
    scanf("%lf", &c);

    double d = pow(b,2) - 4*a*c;
    if (d >= 0) {
        double m,n;
        m = (-b + pow(d,0.5))/(2*a);
        n = (-b - pow(d,0.5))/(2*a);
        printf("The roots of the equation %.2lfx^2 + %.2lfx + %.2lf =0 are \n %.2lf, %.2lf. \n", a,b,c,m,n);
    } else {
        printf("Imaginary Roots :( \n");
    }
}

```

```
    return 0;
}
```

Enter the first coeff. : 2
Enter the second coeff. : 6
Enter the third coeff. : 4
The roots of the equation $2.00x^2 + 6.00x + 4.00 = 0$ are
-1.00, -2.00.

13. To calculate the electric bill using following conditions:

```
#include<stdio.h>
int main() {
    int a;
    printf("ENter the number of units :");
    scanf("%d", &a);
    if (a <= 100) {
        printf("Amount = Rs. 100 \n");
    } else if ( 100 < a && a <= 150) {
        printf("Amount = Rs. %d \n", a*1);
    } else if ( 150 < a && a <= 200) {
        printf("Amount = Rs. %d \n", a*2);
    } else if ( 200 < a) {
        printf("Amount = Rs. %d \n", a*3);
    }
    return 0;
}
```

ENter a number :331
Amount = Rs. 993

14. Print 1 to 100

```
#include<stdio.h>
int main() {
    for (int i = 1; i < 101; i++) {
        printf("%d \n", i);
    }
    return 0;
}
```

15. Print 100 to 1.

```
#include<stdio.h>
int main() {
```

```

    for (int i = 100; i > 0; i--) {
        printf("%d \n", i);
    }
    return 0;
}

```

16. Find Factorial of a number

```

#include <stdio.h>
int fact(int n);
int main() {
    int n;
    printf("Number? : ");
    scanf("%d", &n);
    printf("%d! = %d \n", n ,fact(n));
    return 0;
}
int fact(int n) {
    int m = 1;
    for (int i = 2; i < n+1; i++) {
        m = m * i;
    }
    return m;
}

```

```

Number? : 5
5! = 120

```

17. Find sum of n natural numbers

```

#include<stdio.h>
int main() {
    int n;
    printf("N = ? :");
    scanf("%d", &n);
    printf("Sum is : %d \n", (n*(n+1))/2);
    return 0;
}

```

```

N = ? :5
Sum is : 15

```

18. Check whether a number is Prime or not.

```

#include<stdio.h>
#include<math.h>

int main() {
    printf("Number : ");
    int n;
    //float t = pow(n,0.5);
    int check = 0;
    scanf("%d", &n);
    if (n == 1) {
        check = 2;
        goto printresult;
    } else if ( n == 2) {
        check = 1;
        goto printresult;
    }
    for(int i = 2; i < n; i++){
        if (n % i == 0) {
            check = 0;
            break;
        } else if (n % i != 0) {check = 1;}
    }
    printresult:
    if (check == 1) {
        printf("%d is a prime \n", n);
    } else if (check == 0) { printf("%d is composite \n", n);}
    else if (check == 2) { printf("%d is neither \n", n);}
    return 0;
}

```

```

Number : 3
3 is a prime

```

19. Input a number n and find:

1. Fibonacci series up to n
2. The nth Fibonacci number
3. First n Fibonacci numbers

```

#include<stdio.h>
int main() {
    int a,b,c,n;
    a = 0;
    b = 1;
    printf("i. Enter required n : ");
    scanf("%d",&n);
    printf("Fibonacci series upto n : \n");
}

```

```

printf("0 \n");
printf("%d \n", (n > 1) ? 1 : 0);
c = a + b;
while(c<n+1) {
    printf("%d \n", c);
    a = b;
    b = c;
    c = a + b;
}
printf("\n");
test:
printf("ii. Enter required n : ");
scanf("%d",&n);
printf("nth Fibonacci number : ");
a = 0;
b = 1;
c = a + b;
if (n == 1) {printf("%d \n", 0);}
else if (n == 2) {printf("%d \n", 1);}
int i = 3;
while (i < n + 1) {
    a = b;
    b = c;
    c = a + b;
    i = i + 1;
}
printf("%d \n", c);
a = 1;
b = 1;
c = a + b;
printf("iii. First n Fabinacci numbers : \n");
if (n >= 1) {printf("%d \n", 1);}
if (n >= 2) {printf("%d \n", 1);}
i = 3;
while (i < n + 1) {
    a = b;
    b = c;
    c = a + b;
    printf("%d \n", c);
    i = i + 1;
}
return 0;
}

```

```

i. Enter required n : 10
Fibonacci series upto n :
0
1
1

```

2
3
5
8

ii. Enter required n : 5
nth Fibonacci number : 5
iii. First n Fibonacci numbers :
1
1
2
3
5

20. Check whether a number is Perfect or not (A number is considered to be a Perfect number if sum of factors of the number is equal to itself).

```
#include<stdio.h>
int main() {
    int n;
    printf("Enter number to check: ");
    scanf("%d", &n);
    int i = 1;
    int sum = 0;
    while (i < n) {
        sum = sum + ((n%i == 0) ? i : 0);
        i = i + 1;
    }
    if (sum == n) {
        printf("%d is a Perfect number.", n);
    } else {
        printf("%d is not a Perfect number.", n);
    }
}
```

Enter number to check: 28
28 is a Perfect number.

21. Check whether two numbers are co-prime or not.

```
#include<stdio.h>
int main() {
    int a;
    int b;
    int r, q;
    int x = 1;
    int n, m;
```

```

printf("Enter first number: ");
scanf("%d", &a);
printf("Enter second number: ");
scanf("%d", &b);
n = (a > b)? a : b;
m = (a < b)? a : b;
while (1 == 1) {
    if (m != 0) {
        r = n % m;
        q = n/m;
    } else {
        printf("Zero divison...");
        break;
    }
    printf("%d = %d x %d + %d \n", n,m,q,r);
    if ( r == 1) {m = r; break;}
    else if ( r == 0) {m = 0; break;}
    else {n = m; m = r;}
}
if (m == 1) {printf("Coprime \n");}
else if (m == 0) {printf("NOT Coprime \n");}
else {printf("NOT Coprime \n");}
return 0;
}

```

```

Enter first number: 25
Enter second number: 7
25 = 7 x 3 + 4
7 = 4 x 1 + 3
4 = 3 x 1 + 1
Coprime

```

22. According to the Gold Bach conjecture, every even number greater than two is the sum of two prime numbers. Input an even number and decompose it into two primes.

```

#include<stdio.h>
int isprime(int n);
int main() {
    int n;
    printf("Enter the number to check: ");
    scanf("%d", &n);
    if(n%2 == 0) {
        int i = 3;
        while(i < n) {
            if (isprime(i)) {
                if (isprime(n - i)) {

```

```

        printf("%d = %d + %d \n", n, i, n-i);
        break;
    }
}
i = i + 1;
}
}
}

int isprime(int n) {
    int check = 0;
    for(int i = 2; i < n; i++){
        if (n % i == 0) {
            check = 0;
            break;
        } else if (n % i != 0) {check = 1;}
    }
    if (check == 1) {
        return 1;
    } else if (check == 0) {return 0;}
}

```

Enter the number to check: 52
52 = 5 + 47

23. Find the sum of the digits of a number

```

#include<stdio.h>
int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d",&n);
    int sum = 0;
    while (n != 0) {
        sum = sum + (n%10);
        n = (n - (n%10))/10;
    }
    printf("Sum of digits = %d", sum);
    return 0;
}

```

Enter a number: 2536
Sum of digits = 16

24. Find the Reverse of a number

```

#include<stdio.h>
int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d",&n);
    int rev = n%10;
    n = (n - (n%10))/10;
    while (n != 0) {
        rev = rev*10 + (n%10);
        n = (n - (n%10))/10;
    }
    printf("Reverse of number = %d", rev);
    return 0;
}

```

Enter a number: 12345
Reverse of number = 54321

25. Check whether a number is Armstrong or not (A number is considered to be an Armstrong number if the sum of cubes of its digits is equal to the number itself)

```

#include<stdio.h>
int main() {
    int n;
    printf("Enter a 3 digit number: ");
    scanf("%d",&n);
    int m = n;
    int sum = 0;
    while (n != 0) {
        sum = sum + (int)pow((n%10),3);
        n = (n - (n%10))/10;
    }
    if (sum == m) {
        printf("%d is a Armstrong Number.\n",m);
    } else {
        printf("%d is not a Armstrong Number.\n",m);
    }
    return 0;
}

```

Enter a 3 digit number: 153
153 is a Armstrong Number.

26. Check whether a number is Krishnamurti or not (A number is considered to be a Krishnamurti number if the sum of factorials of its digits is equal to the number itself)

```

#include<stdio.h>
int fact(int n);
int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d",&n);
    int m = n;
    int sum = 0;
    while (n != 0) {
        sum = sum + fact(n%10);
        n = (n - (n%10))/10;
    }
    if (sum == m) {
        printf("%d is a Krishnamurty Number.\n",m);
    } else {
        printf("%d is not a Krishnamurty Number.\n",m);
    }
    return 0;
}
int fact(int n) {
    int m = 1;
    for (int i = 2; i < n+1; i++) {
        m = m * i;
    }
    return m;
}

```

Enter a number: 145
145 is a Krishnamurty Number.

27. Print the following pattern

```

#include <stdio.h>

int printpattern1() {
    for (int i = 0; i < 6; i++) {
        for (int j = 0; j < i + 1; j++) {
            printf("* ");
        }
        printf("\n");
    }
}

void printpattern2() {
    for (int i = 0; i < 6; i++) {
        for (int j = 0; j < i; j++) {
            printf(" ");
        }
    }
}

```

```

    }
    for (int j = 5; j > i; j--) {
        printf("* ");
    }
    printf("\n");
}
}

void printpattern3() {
    for (int i = 0; i < 6; i++) {
        for (int j = 6; j > i + 1; j--) {
            printf(" ");
        }

        for (int j = 0; j < i + 1; j++) {
            printf("* ");
        }
        printf("\n");
    }
}

void printpattern4() {
    for (int i = 0; i < 6; i++) {
        for (int k = 5; k > i; k--) {
            printf(" ");
        }
        for (int j = 0; j < i+1; j++) {
            printf("* ");
        }
        printf("\n");
    }
}

int main() {
    printpattern1(); printf("\n");
    printpattern2(); printf("\n");
    printpattern3(); printf("\n");
    printpattern4(); printf("\n");
}

```

```

*
* *
* * *
* * * *
* * * * *
* * * * * *

* * * * *
* * * *

```

```

* * *
* *
*

      *
    * *
  * * *
* * * *
* * * * *
* * * * * *

      *
    * *
  * * *
* * * *
* * * * *
* * * * * *

```

28. Print the following pattern

```

#include<stdio.h>
int main() {
    for ( int i = 1; i < 7; i++) {
        for ( int j = 1; j <= i; j++) {
            printf("%d ", j);
        }
        printf("\n");
    } printf("\n");
    for ( int i = 1; i < 7; i++) {
        for ( int j = 1; j <= i; j++) {
            printf("%d ", i);
        }
        printf("\n");
    }
}

```

```

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6

```

```

1
2 2
3 3 3
4 4 4 4

```

```
5 5 5 5 5
6 6 6 6 6 6
```

29. Print the following letters with 'star'(*)

```
#include<stdio.h>
int main() {
    int n;
    printf("Please enter a number: ");
    scanf("%d", &n);
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            printf("%s", ((i == j || j == (n-i-1)) ? "*" : " "));
        }
        printf("\n");
    }
    return 0;
}
```

Please enter a number: 5

```
*   *
* *
 *
* *
*   *
```

```
#include<stdio.h>
int main() {
    int n;
    printf("Please enter a number: ");
    scanf("%d", &n);
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            printf("%s", ((j == 0 || j == (n-1) || j == i) ? "*" : "
"));
        }
        printf("\n");
    }
    return 0;
}
```

Please enter a number: 5

```
*       *
* *     *
*   *   *
```

```
*      * *
*      *
```

```
#include<stdio.h>
int main() {
    int n;
    printf("Please enter a number: ");
    scanf("%d", &n);
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            printf("%s", (((i < (n + (n%2))/2) && (i == j || j == (n-i-1))) || ((i >= (n + (n%2))/2) && (j+1==(n + (n%2))/2))) ? "*" : " "));
        }
        printf("\n");
    }
    return 0;
}
```

Please enter a number: 5

```
*      *
* *
*
*
*
```

```
#include<stdio.h>
int main() {
    int n;
    printf("Please enter a number: ");
    scanf("%d", &n);
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            printf("%s", ((i == 0 || i == n-1 || j == (n-i-1)) ? "*" : " "));
        }
        printf("\n");
    }
    return 0;
}
```

Please enter a number: 5

```
* * * * *
      *
     *
```

```
*  
* * * * *
```

30. Print the following order (odd positions- Fibonacci Series and even position- prime numbers)

31. 2 1 3 2 5 3 7 5 11 8 13

```
#include<stdio.h>  
  
int pseries(int i,int n);  
int isprime(int n);  
int fseries(int i,int a,int b);  
  
int pcount, fcount = 1;  
int fnum1, fnum2, pnum1;  
int main() {  
    int n;  
    pnum1 = 2;  
    fnum1 = 1;  
    fnum2 = 2;  
    printf("Enter n: ");  
    scanf("%d", &n);  
    int i = 0;  
    while (i <= n) {  
        if (i%2) {printf("%d ", fseries(fcount, fnum1, fnum2));}  
        else {printf("%d ", pseries(pcount, pnum1));}  
        i = i + 1;  
    }  
}  
  
int pseries(int i, int n) {  
    int j = n+1;  
    if (i == 0) {  
        pcount = pcount + 1;  
        return 2;  
    }else {  
        while (!isprime(j)) {  
            j = j + 1;  
        }  
        pnum1 = j;  
        pcount = pcount + 1;  
        return pnum1;  
    }  
}  
  
int isprime(int n) {  
    int check = 0;  
    for(int i = 2; i < n; i++){  
        if (n % i == 0) {  
            check = 0;  
        }  
    }  
    if (check == 0) return 0;  
    return 1;  
}
```

```

        break;
    } else if (n % i != 0) {check = 1;}
}
if (check == 1) {
    return 1;
} else if (check == 0) {return 0;}
}
int fseries(int i,int a,int b) {
    if (i == 1) {fcount=fcount + 1; return 1;}
    else if (i == 2) {fcount = fcount + 1; return 2;}
    else {
        int c = a + b;
        fcount = fcount + 1;
        fnum1 = fnum2;
        fnum2 = c;
        return c;
    }
}
}

```

Enter n: 10

2 1 3 2 5 3 7 5 11 8 13

32. Find the sum of 1st n prime numbers.

```

#include<stdio.h>
int pcount = 1;
int pnum = 2;
int pseries(int i, int n);
int isprime(int n);

int main() {
    int n;
    int psum = 2;
    printf("Enter n: ");
    scanf("%d", &n);
    while (pcount < n) {
        pseries(pcount, pnum);
        psum = psum + pnum;
    }
    printf("Sum of first %d primes is: %d", n,psum);
    return 0;
}
int pseries(int i, int n) {
    int j = n+1;
    if (i == 0) {
        pcount = pcount + 1;
        return 2;
    }else {

```

```

        while (!isprime(j)) {
            j = j + 1;
        }
        pnum = j;
        pcount = pcount + 1;
    }
}

int isprime(int n) {
    int check = 0;
    for(int i = 2; i < n; i++){
        if (n % i == 0) {
            check = 0;
            break;
        } else if (n % i != 0) {check = 1;}
    }
    if (check == 1) {
        return 1;
    } else if (check == 0) {return 0;}
}

```

Enter n: 5

Sum of first 5 primes is: 28

33. Input a number and check if it a Krishnamurthy number

```

#include<stdio.h>
int fact(int n)
int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d",&n);
    int m = n;
    int sum = 0;
    while (n != 0) {
        sum = sum + fact(n%10);
        n = (n - (n%10))/10;
    }
    if (sum == m) {
        printf("%d is a Krishnamurty Number.\n",m);
    } else {
        printf("%d is not a Krishnamurty Number.\n",m);
    }
    return 0;
}

int fact(int n) {
    int m = 1;
    for (int i = 2; i < n+1; i++) {
        m = m * i;
    }
}

```

```

    }
    return m;
}

```

Enter a number: 145
 145 is a Krishnamurthy Number.

34. Print the Non-Fibonacci series

```

#include<stdio.h>
int main() {
    int a,b,c,n;
    a = 1;
    b = 1;
    printf("Enter required n : ");
    scanf("%d",&n);
    printf("Non Fibonacci series upto n : \n");
    c = a + b;
    while(c < n+1) {
        int i = c + 1;
        a = b;
        b = c;
        c = a + b;
        while(i < c) {
            printf("%d \n", i);
            i = i + 1;
        }
    }
}

```

Enter required n : 10
 Non Fibonacci series upto n :
 4
 6
 7
 9
 10
 11
 12

35. Input the values of two variables n and r and calculate nCr.

```

#include <stdio.h>
int choose(int n,int r);
int fact(int n);
int main() {

```

```

int n,r;
printf("n : ");
scanf("%d", &n);
printf("r : ");
scanf("%d", &r);
printf("%dC%d = %d \n", n , r ,choose(n,r));
return 0;
}
int choose(int n,int r) {
    int c = fact(n)/(fact(n-r)*fact(r));
    return c;
}
int fact(int n) {
    int m = 1;
    for (int i = 2; i < n+1; i++) {
        m = m * i;
    }
    return m;
}

```

```

n : 5
r : 2
5C2 = 10

```

36. To input n numbers from user and display those.

```

#include<stdio.h>
int main() {
    int n;
    printf("How many numbers : ");
    scanf("%d", &n);
    int arr[n];
    for (int i = 0; i < n; i++) {
        printf("Enter number %d: ", i+1);
        scanf("%d", &arr[i]);
    }
    for (int i = 0; i < n; i++) {
        printf("Number_%d= %d; ", i+1 , arr[i]);
    }
}

```

```

How many numbers : 5
Enter number 1: 2
Enter number 2: 3
Enter number 3: 5
Enter number 4: 7

```

Enter number 5: 11

Number_1= 2; Number_2= 3; Number_3= 5; Number_4= 7; Number_5= 11;

37. To input n numbers and find their sum

```
#include<stdio.h>
int main() {
    int n;
    printf("How many numbers : ");
    scanf("%d", &n);
    int sum = 0;
    int arr[n];
    for (int i = 0; i < n; i++) {
        printf("Enter number %d: ", i+1);
        scanf("%d", &arr[i]);
    }
    for (int i = 0; i < n; i++) {
        sum = sum + arr[i]
    }
    printf("SUm = %d", sum);
}
```

How many numbers : 5

Enter number 1: 2

Enter number 2: 3

Enter number 3: 5

Enter number 4: 7

Enter number 5: 11

SUm = 28

38. To find the biggest of n input numbers and find its position

```
#include<stdio.h>
int main() {
    int n;
    printf("How many numbers : ");
    scanf("%d", &n);
    int arr[n];
    for (int i = 0; i < n; i++) {
        printf("Enter number %d: ", i+1);
        scanf("%d", &arr[i]);
    }
    int big = arr[0];
    int index = 0; int temp;
    for (int i = 0; i < n; i++) {
        temp = big;
        big = (arr[i] >= big)? arr[i] : big;
    }
}
```

```

        index = (big == temp)? index : i;
    }
    printf("The largest number is: %d at position: %d", big, index+1);
}

```

```

How many numbers : 5
Enter number 1: 2
Enter number 2: 5
Enter number 3: 3
Enter number 4: 8
Enter number 5: 6
The largest number is: 8 at position: 4

```

39. To input a list of numbers and search a particular number is present or not

```

#include<stdio.h>
int main() {
    int n;
    printf("How many numbers : ");
    scanf("%d", &n);
    int arr[n]; int num;
    int index = 0; int check;
    for (int i = 0; i < n; i++) {
        printf("Enter number %d: ", i+1);
        scanf("%d", &arr[i]);
    }
    printf("Who do you want to search: ");
    scanf("%d", &num);
    check = 0;
    for (int i = 0; i < n; i++) {
        if (arr[i] == num) {check = 1; printf("Found %d at position
%d.\n", num, i+1); break;}
        else {check = 0;}
    }
    if (check == 0) printf("%d not found.", num);
}

```

```

How many numbers : 5
Enter number 1: 1
Enter number 2: 5
Enter number 3: 3
Enter number 4: 2
Enter number 5: 7
Who do you want to search: 2
Found 2 at position 4.

```

40. Do the previous problem using binary search.

```
#include<stdio.h>
int sort(int arr[], int n) {
    int temp;
    for(int i=0; i < n-1; i++) {
        for (int j = 0; j < n-1-i; j++) {
            if (arr[j] > arr[j+1]) {
                temp = arr[j];
                arr[j] = arr[j+1];
                arr[j+1] = temp;
            }
        }
    }
}

int binSearch(int target,int n,int arr[]) {
    int high = n-1;
    int low = 0;
    while (low <= high) {
        int mid = (low + high)/2;
        if (arr[mid] == target) {
            return mid;
        } else if (mid > target) {
            high = mid - 1;
        } else {
            low = mid + 1;
        }
    }
    return -1;
}

int main() {
    int n, target;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    printf("Enter the target to search for: ");
    scanf("%d", &target);
    int arr[n];
    printf("Enter the array elements: \n");
    for (int i = 0; i < n; i++) {
        printf(">>");
        scanf("%d", &arr[i]);
    }
    sort(arr, n);
    int result = binSearch(target, n, arr);
    if (result != -1) {
        printf("%d found at indes %d in sorted array.",target, result+1);
    } else {
        printf("Error 404 !");
    }
}
```

```
    return 0;
}
```

```
Enter the number of elements: 5
Enter the target to search for: 3
Enter the array elements:
>>2
>>5
>>3
>>4
>>1
3 found at index 3 in sorted array.
```

41. Find the factorial of a number using function

```
#include <stdio.h>
int fact(int n);
int main() {
    int n;
    printf("Number? : ");
    scanf("%d", &n);
    printf("%d! = %d \n", n ,fact(n));
    return 0;
}
int fact(int n) {
    int m = 1;
    for (int i = 2; i < n+1; i++) {
        m = m * i;
    }
    return m;
}
```

```
Number? : 5
5! = 120
```

43. Draw the Pascal Triangle

```
#include<stdio.h>
int choose(int n,int r);
int fact(int n);
int main() {
    int n;
    printf("How many lines : ");
    scanf("%d", &n);
    int space = n;
    for ( int i = 0; i < n + 1; i++) {
```

```

        int sp = space - i;
        for (int k = 0; k < sp; k++) {
            printf(" ");
        }
        for (int j = 0; j < i + 1; j++){
            printf("%d ", choose(i,j));
        }
        printf("\n");
    }
    return 0;
}

int choose(int n,int r) {
    int c = fact(n)/(fact(n-r)*fact(r));
    return c;
}

int fact(int n) {
    int m = 1;
    for (int i = 2; i < n+1; i++) {
        m = m * i;
    }
    return m;
}

```

How many lines : 8

```

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1
1 8 28 56 70 56 28 8 1

```

44. Find factorial of a number using recursion

```

#include<stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d",&n);
    int fact = 1;
    for (int i = 1; i < n+1; i++) {
        fact = fact * i;
    }
    printf("%d! = %d", n, fact);
}

```

```
    return 0;
}
```

Enter n: 6
6! = 720[

45. Sum of n natural numbers using recursion

```
#include<stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d",&n);
    int sum = 0;
    for ( int i = 1; i < n+1; i++) {
        sum = sum + i;
    }
    printf("Sum of first %d numbers = %d", n, sum);
    return 0;
}
```

Enter n: 5
Sum of first 5 numbers = 15

46. Find a^b using recursion

```
#include<stdio.h>
int main() {
    int a,b;
    printf("Enter a: ");
    scanf("%d",&a);
    printf("Enter b: ");
    scanf("%d",&b);
    int power = 1;
    for ( int i = 0; i < b; i++) {
        power = power*a;
    }
    printf("%d ^ %d = %d", a,b, power);
    return 0;
}
```

Enter a: 5
Enter b: 5
5 ^ 5 = 3125

47. Print Fibonacci series using recursion

```
#include<stdio.h>
int main() {
    int n;
    printf("Enter n : ");
    scanf("%d", &n);
    int arr[n];
    arr[0] = 1; arr[1] = 1;
    for ( int i = 2; i <=n ; i++) {
        arr[i] = arr[i-1] + arr[i-2];
    }
    for (int i = 0; i < n; i++) {
        printf("%d \n", arr[i]);
    }
}
```

```
Enter n : 5
1
1
2
3
5
```

48. Input two numbers and find their GCD using recursion.

```
#include<stdio.h>
int main() {
    int a,b;
    int n,m;
    int r,q;
    printf("Enter first number: ");
    scanf("%d", &a);
    printf("Enter second number: ");
    scanf("%d", &b);
    n = (a > b)? a : b;
    m = (a < b)? a : b;
    while (1 == 1) {
        if (m != 0) {
            r = n % m;
            q = n/m;
        } else {
            printf("Zero divison... \n");
            goto endme;
            break;
        }
        printf("%d = %d x %d + %d \n", n,m,q,r);
    }
```

```

        if ( r == 1) {m = r; break;}
        else if (r == 0) {break;}
        else {n = m; m = r;}
    }
    printf("gcd(%d,%d) = %d \n",a,b,m);
endme:
    return 0;
}

```

Enter first number: 72
Enter second number: 16
72 = 16 x 4 + 8
16 = 8 x 2 + 0
gcd(72,16) = 8

49. Input two strings and concatenate them.

```

#include<stdio.h>
#include<string.h>
int main() {
    char str1[100], str2[100];
    printf("Enter first string :");
    fgets(str1, sizeof(str1), stdin);
    printf("Enter second string :");
    fgets(str2, sizeof(str2), stdin);
    int n = strlen(str1) + strlen(str2);
    char str3[n];
    int count;
    for (int i=0; str1[i] != '\0'; i++) {
        count = i;
        if (str1[i] != '\n' || str1[i] != '\n') {
            str3[i] = str1[i];
        }
    }
    for (int i=0; str2[i] != '\0'; i++) {
        if (str2[i] != '\n' || str2[i] != '\n') {
            str3[i + count] = str2[i];
        }
    }
    for (int i = 0; str3[i] != '\0'; i++) {
        printf("%c", str3[i]);
    }
    return 0;
}

```

Enter first string :String concate
Enter second string :nation in C programming

50. Input a string and reverse it.

```
#include<stdio.h>
#include<string.h>
int main() {
    char n[100];
    printf("Enter the string : ");
    scanf("%99[^\n]", n);
    int i = strlen(n) - 1;
    while(i >= 0) {
        printf("%c",n[i]);
        i = i -1;
    }
}
```

Enter the string : Avada Kedavra
arvadeK adavA

51. Enter a sentence and find number of vowels, consonants, spaces and special characters

```
#include <stdio.h>
#include <ctype.h>
#include<string.h>

int main() {
    char str[200];
    int v = 0, c = 0, s = 0, sp = 0;

    printf("Enter a sentence: ");
    fgets(str, sizeof(str), stdin);

    for (int i = 0; str[i] != '\0'; i++) {
        char ch = tolower(str[i]);
        if (isalpha(ch)) {
            if (strchr("aeiou",ch))
                v++;
            else
                c++;
        }
        else if (isspace(ch)) {
            if (ch != '\n') s++;
        }
        else if (ch != '\n' && ch != '\r') {
            sp++;
        }
    }
}
```

```

    }
    printf("\nVowels: %d", v);
    printf("\nConsonants: %d", c);
    printf("\nSpaces: %d", s);
    printf("\nSpecial Characters: %d\n", sp);

    return 0;
}

```

Enter a sentence: hello world

Vowels: 3

Consonants: 7

Spaces: 1

Special Characters: 0

52. Check a string palindrome or not.

```

#include <stdio.h>
#include <string.h>

int isPalindrome(char str[]) {
    int low = 0;
    int high = strlen(str) - 1;

    while (low < high) {
        if (str[low] != str[high]) {
            return 0;
        }
        low++;
        high--;
    }
    return 1;
}

int main() {
    char word[100];
    printf("Enter a string: ");
    scanf("%s", word);

    if (isPalindrome(word)) {
        printf("%s is a palindrome.\n", word);
    } else {
        printf("%s is not a palindrome.\n", word);
    }
}

```

```
    return 0;
}
```

Enter a string: malayalam
malayalam is a palindrome.

53. Find reverse of a string.

```
#include<stdio.h>
#include<string.h>
int main() {
    char n[100];
    printf("Enter the string : ");
    scanf("%99[^\n]", n);
    int i = strlen(n) - 1;
    while(i >= 0) {
        printf("%c", n[i]);
        i = i - 1;
    }
}
```

Enter the string : helloWorld
dlroWolleh